

End_Term_Q1&2_Codes

Question 1

```
# albums = {}
```

```
# with open("albums.txt") as file:
```

```
# for line in file:
```

```
# data = line.strip().split(";")
```

```
# name = data[1]
```

```
# album = data[0]
```

```
#
```

```
# if name in albums:
```

```
# albums[name] += 1
```

```
# else:
```

```
# albums[name] = 1
```

```
#
```

```
# for key, value in sorted(albums.items(), key=lambda x: (-x[1], x[0])):
```

```
# print(f"{key}: {value} album(s) available")
```

```
# Question 2
```

```
# import sys
```

```
#
```

```
# devices = {}
```

```
# for line in sys.stdin:
```

```
# data = line.strip().split(":")
```

```
# name = data[0]
```

```
# device = data[1].split(",")
```

```
# num_of_devices = len(device)
```

```
#
```

```
# if name in devices:
```

```
# devices[name] += num_of_devices
```

```
# else:
```

```
# devices[name] = num_of_devices
```

```
#
```

```
# for key, value in sorted(devices.items(), key=lambda x: (x[1], x[0])):
```

```
# print(f"{key}: {value} product(s)")
```

```
# Question 3
```

```
# devices = {}
```

```
# with open("devices.txt") as file:
```

```
# for line in file:
```

```
# data = line.strip().split(",")
```

```
# device = data[1]
```

```
# price = int(data[3])
```

```
#
```

```
# if device in devices:
```

```
# devices[device] += price
```

```
# else:
```

```
# devices[device] = price
```

```
# for key, value in sorted(devices.items(), key=lambda x: (-x[1], x[0])):
```

```
# print(f"{key} ({value} EUR)")
```

```
# Question 4
```

```
# import sys
```

```
#
```

```
# attractions = {}
```

```
# for line in sys.stdin:
```

```
# data = line.strip().split(":")
```

```
# city_name = data[0]
```

```
# attraction = data[1].split(",")
```

```
# num_of_attraction = len(attraction)
```

```
#
```

```
# if city_name in attractions:
```

```
# attractions[city_name] += num_of_attraction
```

```
# else:
```

```
# attractions[city_name] = num_of_attraction
```

```
# for key, value in sorted(attractions.items(), key=lambda x: (x[1], x[0])):
```

```
# print(f"{key}: {value} attraction(s)")
```

```
# Question 5
```

```
# bicycles = {}
```

```
# with open("bicycles.txt") as file:
```

```
# for line in file:
```

```
# data = line.strip().split(":")
```

```
# type = data[1]
```

```
# price = int(data[3])
```

```
#
```

```
# if type in bicycles:
```

```
# bicycles[type] += price
```

```
# else:
```

```
# bicycles[type] = price
```

```
# for key, value in sorted(bicycles.items(), key=lambda x: (-x[1], x[0])):
```

```
# print(f"{key} ({value} HUF)")
```

```
# Question 6
```

```
# import sys
```

```
#
```

```
# fruits = {}
```

```
# for line in sys.stdin:
```

```
# data = line.strip().split(":")
```

```
# name = data[0]
```

```
# fruit = data[1].split(",")
```

```
#
```

```
# for i in fruit:
```

```
# if i in fruits:
```

```
# fruits[i] += 1
```

```
# else:
```

```
# fruits[i] = 1
```

```
# for key, value in sorted(fruits.items(), key=lambda x: (x[0])):
```

```
# print(f"{key}: {value} child")
```

```
# Question 7
```

```
# countries = {}
```

```
# with open("country.txt") as file:
```

```
# for line in file:
```

```
# data = line.strip().split(";")
```

```
# country = data[0]
```

```
# population = int(data[2])
```

```
#
```

```
# if country in countries:
```

```
# countries[country] += population
```

```
# else:
```

```
# countries[country] = population
```

```
# for key, value in sorted(countries.items(), key=lambda x: (-x[1], x[0])):
```

```
# print(f"{key} ({value} people)")
```

```
# Question 8
```

```
# import sys
```

```
#
```

```
# hobbies = {}
```

```
# for line in sys.stdin:
```

```
# data = line.strip().split(":")
```

```
# name = data[0]
```

```
# hobby = data[1].split(",")
```

```
#
```

```
# for i in hobby:
```

```
# if i in hobbies:
```

```
# hobbies[i] += 1
```

```
# else:
```

```
# hobbies[i] = 1
```

```
#
```

```
# for key, value in sorted(hobbies.items(), key=lambda x: (x[0])):
```

```
# print(f"{key}: {value} persons")
```

```
# Question 9
```

```
# clothes = {}
```

```
# with open("clothes.txt") as file:
```

```
# for line in file:
```

```
# data = line.strip().split(";")
```

```
# type = data[0]
```

```
# price = int(data[3])
```

```
#
```

```
# if type in clothes:
```

```
# clothes[type] += price
```

```
# else:
```

```
# clothes[type] = price
```

```
#
```

```
# for key, value in sorted(clothes.items(), key=lambda x: (-x[1], x[0])):
```

```
# print(f"{key} ({value} EUR)")
```

```
# Question 10
```

```
# import sys
```

```
#
```

```
# subjects = {}
```

```
# for line in sys.stdin:
```

```
# data = line.strip().split(":")
```

```
# name = data[0]
```

```
# subject = data[1].split(",")
```

```
#
```

```
# for i in subject:
```

```
# if i in subjects:
```

```
# subjects[i] += 1
```

```
# else:
```

```
# subjects[i] = 1
```

```
#
```

```
# for key, value in sorted(subjects.items(), key=lambda x: (-x[1], x[0])):
```

```
# print(f"{key}: {value} student (s)")
```

Question 11

books = {}

with open("books.txt") as file:

for line in file:

data = line.strip().split(";")

author = data[1]

book = data[0]

#

```
# if author in books:
```

```
# books[author] += 1
```

```
# else:
```

```
# books[author] = 1
```

```
#
```

```
# for key, value in sorted(books.items(), key=lambda x: (-x[1], x[0])):
```

```
# print(f"{key}: {value} book(s) available")
```

```
# Question 12
```

```
# import sys
```

```
#
```

```
# projects = {}
```

```
# for line in sys.stdin:
```

```
# data = line.strip().split(":")
```

```
# name = data[0]
```

```
# project = data[1].split(",")
```

```
# num_of_projects = len(project)
```

```
#
```

```
# if name in projects:
```

```
# projects[name] += num_of_projects
```

```
# else:
```

```
# projects[name] = num_of_projects
```

```
#
```

```
# for key, value in sorted(projects.items(), key=lambda x: (-x[1], x[0])):
```

```
# print(f"{key}: {value} project(s)")
```

```
# Question 13
```

```
# clothes = {}
```

```
# with open("clothes2.txt") as file:
```

```
# for line in file:
```

```
# data = line.strip().split(";")
```

```
# type = data[0]
```

```
# price = int(data[3])
```

```
# pieces = int(data[2])
```

```
#
```

```
# if type in clothes:
```

```
# clothes[type] += int(price / pieces)
```

```
# else:
```

```
# clothes[type] = int(price / pieces)
```

```
#
```

```
# for key, value in sorted(clothes.items(), key=lambda x: (-x[1], x[0])):
```

```
# print(f"{key} ({value} EUR)")
```

```
# Question 14
```

```
# import sys
```

```
#
```

```
# subjects = {}
```

```
# for line in sys.stdin:
```

```
# data = line.strip().split(":")
```

```
# name = data[0]
```

```
# subject = data[1].split(",")
```

```
#
```

```
# for i in subject:
```

```
# if i in subjects:
```

```
# subjects[i] += 1
```

```
# else:
```

```
# subjects[i] = 1
```

```
#
```

```
# for key, value in sorted(subjects.items(), key=lambda x: (-x[1], x[0])):
```

```
# print(f"{key}: {value} student(s)")
```